

A Prototype Intelligent Agent Based DSS for Quadratic Portfolio Optimization Calculations with Case Based Reasoning

Edward Falconer^{*}
University of Paisley

Abel Usoro
University of Paisley

Mark Stansfield
University of Paisley

Brian Lees
University of Paisley

Abstract

A variety of intelligent agent systems have assisted portfolio management decisions. The majority of these systems have focused on analysis methods, such as fundamental analysis, technical analysis or analysis of trader behavior. According to Clarke et al, because of the efficient markets hypothesis, portfolio managers should limit the time that they spend on analysis of this type and instead concentrate on other aspects of portfolio management, such as diversification, asset allocation or cost management. This study investigates the application of intelligent agents to strategic asset allocation. Traditionally, strategic asset allocation is calculated using tools for quadratic optimization and determining an investor's utility towards risk. Associated with strategic asset allocation are many complexities, such as the distributed nature of problems that make a solution difficult to implement. This paper outlines results of an evaluation of whether intelligent agent and case based reasoning could overcome these complexities. From a review of literature there does not appear to have been a study that has considered this problem in the way proposed in this paper.

1. Introduction

The findings from the efficient markets hypothesis have led academics and practitioners to divide investment management into two primary types: i) active investment management, where an investment manager uses available information and forecasting techniques to seek better performance than a diversified portfolio; and ii) passive investment management, where an investment manager has minimal expectations about the returns that can be achieved and relies instead on diversification to match the performance of a market index [1], [2]. Although each of these approaches has benefits, there has been

strong evidence to support the claim that the profitability of active investing is minimal, especially after transaction costs, and therefore passive investing is more profitable [1], [3], [4]. One of the implications of choosing a passive investment approach is that portfolio managers must concentrate on other strategies for improving on the return from their portfolio [1]. Examples of these strategies include cost management, diversification and asset allocation strategies. This research concentrates on one approach to asset allocation – strategic asset allocation.

Asset allocation is a process whereby an overall sum of cash is separated into different portions with each portion then allocated to a different category of asset class. According to Sharpe [5], [6] each asset class should represent a capitalisation-weighted portfolio of securities in order to mimic the return variation created by different weights of asset class in the returns of the portfolio under evaluation. The optimal strategic asset allocation can be determined in one of two ways: by maximizing return for a given risk level or alternatively, minimizing risk for a particular return objective [7], [8], [9], [10]. A strategic asset allocation decision is made up of a quadratic component and a utility component [7]. The quadratic component is concerned with optimization calculations, while the utility component is concerned with evaluating an investor's acceptance of risk. Moreno *et al.*, [11] described Markowitz's mean variance approach as the classical model for risk forecasting.

This research considers the application of intelligent agents and case based reasoning (CBR) to strategic asset allocation. CBR has been used before for investment planning, technical portfolio analysis (analysing price graphs) and fundamental portfolio analysis (analysing company announcements) (see [12], [13], [14], [15], [16], [17] for some examples). CBR has been used to support portfolio management decisions where a problem is distributed across

^{*} Corresponding author. E-mail: e.falconer@sdlsolutions.com

multiple components. Intelligent agents have also been used for aiding portfolio management decisions [18]. Luo *et al.*, [19] developed an agent based trading system that performed statistical analysis for trading decisions. Decker and Sycara [14] and Tseng and Gymtrasiewicz [20] detailed the importance of an investor's tolerance for risk and their tax situation. This was also highlighted by Davis *et al.*, [13], Davis and Luo [21] and Luo *et al.*, [19]. In agent literature the determination of an investor's utility or support of quadratic programming does not appear to have been considered. This research assesses the effectiveness of intelligent agents and CBR to this problem. This paper concentrates on the quadratic program calculations under mean variance efficiency (MVE). Utility theory will be considered in a subsequent paper.

The rest of this paper provides: (i) a background to the research problem and overview of the data preparation steps; (ii) an overview of agent and CBR characteristics; (iii) an outline of the agent architecture; (iv) application of case based reasoning to the problem and preliminary results; (v) conclusions and future directions.

2. Background and Data Preparation

The aim of quadratic programming in the portfolio selection problem is to calculate the amount that should be invested in different securities based on covariance and expected returns. Markowitz [22], [23], [24] defined the following objective function to determine how to maximise returns or minimise risk subject to an individual's risk preferences.

$$\begin{aligned} & \min(\max) \text{imise } \sum \mu_i x_i - p \sum \sum \sigma_{ij} x_i x_j \\ & \text{subject to:} \\ & \sum x_i = 1 \text{ \&} \\ & x_i \geq 0 \text{ for all } i. \end{aligned}$$

In the expression, p represents the investor's aversion to risk, μ_i represents the return on the i th security, x_i represents the amount invested in the i th security and σ_{ij} refers to the covariance of the i th and j th security.

The simplest way of implementing a solution under quadratic and linear constraints is to substitute the investor's risk aversion for a value representing the minimum portfolio return [25]. This minimum return value then replaces the letter p in the equation and allows the portfolio to be determined based on return and risk. By taking this approach the quadratic programming problem is reduced to a question about which asset classes have been chosen and what is the minimum portfolio return.

The expected return and variance used within standard quadratic calculation are determined using historic share price information. This study is using global asset prices. The global strategy differs slightly from the standard asset allocation approach as the focus is on index prices. These indices reflect the price of all equities within a specified global market. In contrast, asset allocation approaches usually focus on broader asset allocation classes, such as real estate, stocks, bonds and cash. Despite the differences there are many advantages to adopting a global asset allocation approach (for in-depth discussions refer to [26], [27], [28]). These advantages include the benefits of global asset coverage and the general stability of global markets [26], [27]. There is also the advantage that comes from having daily prices available.

Historical data was obtained from Morgan Stanley Corporate International (MSCI©) for a variety of global indices. This selection of assets provided broad coverage across developing and established markets. Although the assets are limited to those specified and have only included share assets, the model could easily be extended to include additional global asset classes, cash assets, or bond assets. Ten years' worth of MSCI global index prices were used to calculate the mean returns on each asset class. Variances, standard deviations, covariances and correlation coefficients were also calculated for each asset class. The underlying data is available from the author by request.

3. Agent & CBR Characteristics

From an evaluation of the asset allocation process a number of complexities were observed. These complexities had the potential to present obstacles to the effectiveness of a DSS designed to assist with strategic asset allocation decisions. For instance: (i) the asset allocation task was of a distributed nature; this meant that many different factors were contributing to each decision; (ii) some tasks required input from other tasks to arrive at a conclusion; thereby necessitating the need to share data and control the flow of information between tasks; (iii) there was a chance that the same set of data inputs could be used multiple times; this meant there was a potential for duplication of tasks; (iv) there was not one solution that could be applied to every set of circumstances; this meant that each decision could be either unique or similar to a previous decision; (iv) the decision involved input from a user; and this user could require more or less assistance depending on their preferences.

In the light of these observations a review was undertaken of available technologies. During this

review intelligent agent and case based reasoning were identified as offering the types of attributes that could be effective in dealing with complexities of the types identified. The following section outlines some of these attributes.

3.1 Intelligent Agents

Intelligent agents are software artefacts that have a number of characteristics that distinguish them from other technologies. In particular, an agent can be given a goal or task which it can complete autonomously. Agents may also carry out tasks or goals in parallel with the existing computer operations and ask for advice from humans if they become “stuck” in a problem. Agents can act proactively and communicate with human users, providing a personalised service to specific users and communicating with other agents in a multi-agent environment. Intelligent agents have the following capabilities that suggest they may be appropriate to this situation. More specifically, intelligent agents can:

React to change: Intelligent agents can interact with, and react to, a user’s preferences. This means that software can be designed to recognise a user’s actions and modify the user/agent interaction to ensure an approach appropriate to that user is adopted.

Organise data flow: Intelligent agents can incorporate a separate control component to regulate the flow of information between distributed tasks. This control component organises the flow of information that passes between the various parts of the problem and ensures that a conclusion is reached.

Construct knowledge bases: To build an autonomous system it is necessary to acquire knowledge from a variety of knowledge sources: human expertise, recorded knowledge of previous successful problem solutions. Intelligent agents take this collection of knowledge and form the system’s knowledge by suitably structuring the data.

Learn from experience: The overall aim is usually for the intelligent agent system to support the problem solving activities of the human expert. Over time it may be necessary to maintain the knowledge base, by identifying and correcting any inconsistencies, and by filling in any perceived gaps through the acquisition and inclusion of additional knowledge. Intelligent agents also demonstrate this feature.

3.2. Case Based Reasoning

CBR is a methodology that can be used to allow software to learn from its experiences. Case based reasoning subdivides a problem into a series of tasks

and then combines these to form a case. Each case is then compared with previous cases to determine whether new experiences can be learned. Fig 1 illustrates a general CBR cycle [29]. Four tasks are included: *Retrieve task* – to retrieve similar cases, *Reuse task* – to reuse information and knowledge from previous cases to solve a problem, *Revise task* – where the problem is revised in line with previous experiences, *Retain task* – where experience is retained that could be used for future cases. Using this general cycle a software program can learn from its experiences.

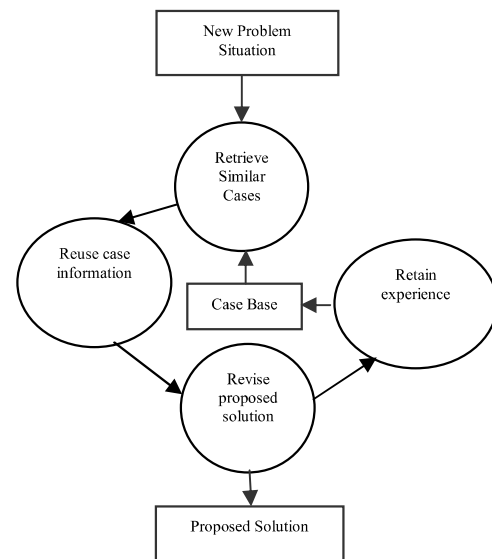


Fig 1: General CBR Program Cycle

CBR was found to offer the types of capabilities that may make it appropriate for dealing with the complexities of the asset allocation decision. CBR technologies:

Organise distributed tasks: CBR systems gather case data from distributed tasks and join the outputs of these tasks together, to formulate a combined state. In this way a distributed problem can be resolved, with each distributed solution contributing to all future solutions.

Subdivide tasks into categories: CBR systems subdivide the solution into different parts and then pass requests for analysis onto other components. By initiating this task division CBR software can perform distributed tasks at the same time as other tasks are performed and then combine the parts to arrive at a more complete solution.

Evaluate current cases: CBR systems evaluate each set of results to determine whether the current case is new, or whether it matches a previous case. The CBR system then adds new cases and modifications to

existing cases to the case base for the future benefit of the software.

Learn from experience: Each case is made up of a set of responses that are unique to each problem and solution. By comparing current information and approaches with new sets of input CBR programs are able to identify similarities and differences with each new set of inputs. In so doing the CBR system therefore learns from its experiences.

The adoption of a combined agent and CBR based approach is expected to offer a more effective solution than using each of these technologies in isolation.

4. Intelligent Agent Architecture

The agent architecture is designed to separately carry out each task and then combine the results to make an asset allocation decision for the user. The agent architecture is shown in Fig 2. There are five parts to the agent: (i) investor profiler; (ii) asset selector; (iii) control component; (iv) case based reasoning component; and (v) maths calculator. Each component is outlined below.

Investor Profiler: The investor profiler retrieves questions from a questionnaire class linked to a database. The database stores a series of questions which the investor profiler then asks the user. Each question also has a set of responses. The investor profiler asks the user the questions and presents some possible answers. The investor then chooses one of the answers from those presented. The investor profiler then writes these results to the database and sends the investor's risk profile to the control component.

Asset Selector: The asset selector component is tasked with choosing assets that match the profile of the investor. The choice of asset is evaluated based on the information passed by the control component which initially comes from the investor profiler. The asset selector retrieves input in the form of asset prices from asset pricing files. The asset selector component also communicates the results to the user.

Control Component: The control component is tasked with controlling the flow of information between the agent's various parts. The control component is used primarily to control the flow of information within the agent environment. The control component is tasked with passing information for processing and passing processed information around the agent environment.

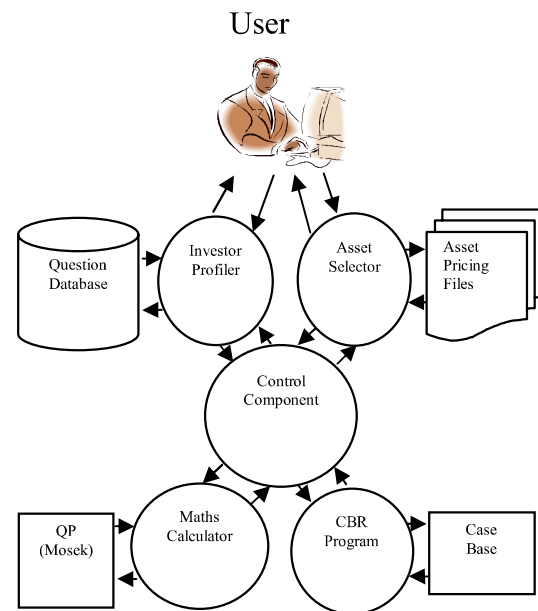


Fig 2: Prototype Architecture for Asset Allocation

The Maths Calculator: The maths calculator component interacts with the quadratic program and modifies the constraints, number of variables and other inputs, to allow the quadratic program to make its calculations. The maths calculator determines the inputs and reads the solution. It plays the part of a driver program for information being passed to the quadratic program (QP) program. Mosek was used to perform the quadratic calculations.

The Case Based Reasoning component: The CBR component interacts with the control component and tracks information received from the different parts of the agent architecture. It uses inferences around the inputs that have been received previously to determine whether information can be used again, or whether parts of previous problems can be used again. The CBR sends its results to the control component.

The agent program is designed to support the activities associated with the asset allocation task. The remainder of this paper outlines preliminary results for the maths calculator and provides a theoretical overview of the types of matching rule that could be used by the CBR component in the completed program.

5. The Quadratic Program and CBR Based Results

The QP used a quadratic function to determine the optimum investment amount to place in each asset class. The following objective function was used to maximise return against risk (from an article by Fylstra [32]).

$$\begin{aligned} &\text{maximise} && c^T x - x^T Q^o x \\ &\text{subject to} && Ax \geq b \\ &&& \Sigma x = 1 \\ &&& x_i \geq 0 \text{ for all } i. \end{aligned}$$

Q^o is a matrix containing covariance for the assets held within the portfolio. c^T is a vector containing returns on each of the assets held within the portfolio. A is also a vector containing the returns on each of the assets held within the portfolio. b is a value that represents the minimum portfolio return. x is the amount to be invested in the i th asset class within the portfolio.

The values associated with the portfolio are then adjusted based on the minimum portfolio return and the asset selector's choice of asset class. Within the equation the constraints ($Ax \geq b$; $\Sigma x = 1$; $x_i \geq 0$ for all i) remained the same, with the values associated with the Q^o , c^T and A , adjusting in line with the input of the problem variables.

Value	Description
Number of variables	The number of asset classes chosen by the asset selector
Number of variables in the matrix A	The same as the number of asset classes chosen by the asset selector
Number of variables in the matrix Q^o	Calculated using the following equation (where N is the number of asset classes chosen by the asset selector): $\frac{N(N+1)}{2}$
Values within the matrix A	The expected return for each asset class
Values for the matrix c^T	The same as the expected return for each asset class
Values within the matrix b	The minimum portfolio return
Values within the matrix Q^o	The covariance for each value N_i, N_j

Table 1: Variables used by the MOSEK tool and determined by the maths calculator

A sample quadratic equation was input into the MOSEK tool which calculated the optimal amount that

should be invested to obtain a minimum portfolio return of 18%. For the purposes of the sample, four countries were chosen for the equation: USA, Canada, Germany and UK. The output of the sample returned by the maths calculator and retrieved from the MOSEK tool is displayed in Table 2.

Country	Variable X	Optimum investment amount
USA	X0	0.289805523
Canada	X1	0.306632754
Germany	X2	0.189994446
UK	X3	0.213567277

Table 2: Optimal results for the quadratic code and sample inputs

The values shown in Table 1 identify the variables used by the maths calculator to make the quadratic calculations. These values were determined from either the data held within the matrices or the chosen asset classes. Each value served a dual purpose: (i) to enable the maths calculator to input the appropriate data to allow the MOSEK tool to calculate the results; and (ii) to allow the CBR program to determine inferences about the data. The size of the three matrices and vectors (Q^o , c^T and A) were increased in line with the number of variables included in the portfolio.

The CBR program also uses the variables identified in Table 1 to develop a matching strategy. The following strategies were identified for the CBR component: (1) match against the same minimum portfolio return; (2) match against the same number of asset classes; (3) match from having one or more of the same asset classes within a portfolio. These matching rules will be used both to retrieve similar cases and also to identify matching cases.

6. Conclusions and Future Direction

This paper has outlined a decision-support system that is based on an intelligent agent. The system uses case-based reasoning to calculate portfolio optimisation under quadratic constraints. These calculations are performed as part of an optimisation of the expected return and the minimisation of risk within a portfolio of broad asset classes.

The results attained from an evaluation of the maths calculator present sufficient evidence to support the decision to use intelligent agents for the problem. At this stage very little work has been carried out into the application of CBR to this problem. However, the

capabilities exhibited by general CBR programs suggest that this may be an appropriate approach to use.

By adopting a combined agent and CBR approach the system can learn from its interactions with the user and also its experiences when evaluating problems.

The next piece of work to be undertaken will involve the development of the additional agent components; in particular, the focus will largely be placed on development of the agent's utility component.

References

- [1]. J. Clarke, G. Mendelker, T. Jandik, *Expert Financial Planning : Investment Strategies from Industry Leaders*, Edited by Robert C Arffa, chapter 10, Wiley and Sons, 2001.
- [2]. E.F. Fama, "Efficient Capital Markets: II", *Journal of Finance*, December, 1991.
- [3]. W. Bernstein, *The Intelligent Asset Allocator: How to Build Your Portfolio to Maximize Returns and Minimize Risk* ; McGraw-Hill Education, 2000.
- [4]. W. Bernstein, *The Four Pillars of Investing: Lessons for Building a Winning Portfolio*, McGraw-Hill Education, 2002.
- [5]. W. Sharpe, "Asset Allocation: Management Style and Performance Measurement", *The Journal of Portfolio Management*, Winter 1992.
- [6]. W. Sharpe, *Budgeting and Monitoring the Risk of Defined Benefit Pension Funds*, <http://www.stanford.edu/~wfs Sharpe/art/q2001/q2001.htm>, 2001.
- [7]. H. Nemati, L.S. Iyer, "Intelligent decision support system for asset allocation", *Americas Conference on information Systems*, 1999.
- [8]. P. Kaplan, "Asset Allocation Models Using the Markowitz Approach", http://www.misp.it/doc/materiali_doc/Curti1-EN.pdf, 1998
- [9]. G.P. Todd, H.M. Markowitz, *Mean-Variance Analysis in Portfolio Choice and Capital Markets*; Frank J. Fabozzi Associates, 2000.
- [10]. F.J. Fabozzi, H.M. Markowitz, *The Theory and Practice of Investment Management* (Frank J. Fabozzi Series) ; John Wiley & Sons, 2002.
- [11]. D. Moreno, P. Marco; I. Olmeda, "Risk forecasting models and optimal portfolio selection", *Applied Economics*, Volume 37, Number 11, 20, June, pp. 1267-1281(15), Routledge, part of the Taylor & Francis Group, 2005.
- [12]. Y. Luo, D. Davis, K. Liu, "Information and Knowledge Exchange in a Multi-Agent System for Stock Trading", *IEEE/IEC Enterprise Networking Applications and Services Conference (EntNet2001)*, Atlanta, USA, July 2001.
- [13]. D. Davis, Y. Luo, K. Liu, "A Multi-Agent Framework for Stock Trading", *World Computing Conference*, Beijing, August 2000.
- [14]. K.S. Decker, K. Sycara, "Intelligent Adaptive Information Agents", *Journal of intelligent Information Systems*, Kluwer Academic Publishers, volume 9, 1997, pp 239 – 260.
- [15]. Y. W. Seo, J. A. Giampapa, and K. Sycara, "Text classification for intelligent agent portfolio management," *First International Conference on Autonomous Agents and Multi-Agent System* 2002.
- [16]. M. Paolucci, Z. Niu, Z. K. Sycara, C. Domashnev, S.R. Owens, M. Van Velsen, "Matchmaking to Support Intelligent Agents for Portfolio Management", In *Proceedings of AAAI (Demo Session)*, 2002.
- [17]. D. Davis, Y. Luo, "Using KADS to Design a Multi-Agent Framework for Stock Trading", *Multi-Conference Event - Agent for E-Business on the Internet*, Las Vegas, June 2001.
- [18]. M. Wooldridge, *An Introduction to Multiagent Systems*, John Wiley and Sons Ltd, 2002.
- [19]. Y. Luo, D. Davis, K. Liu, "A multi-agent system framework for decision making support in stock trading", *The IEEE Network Magazine Special Issue on Enterprise Networking Services*, Volume 16, No 1, Jan/Feb 2002.
- [20]. C. Tseng, P.J. Gmytrasiewicz, "Real Time Decision Support System for Portfolio Management", *35th Annual Hawaii International Conference on System Sciences (HICSS'02)*, Volume 3, January 07 – 10, 2000.
- [21]. D. Davis, Y. Luo, "Using KADS to Design a Multi-Agent Framework for Stock Trading", *Multi-Conference Event - Agent for E-Business on the Internet*, Las Vegas, June 2001.
- [22]. H. Markowitz, "Portfolio Selection, *Journal of Finance*", 7(1): 77-91, 1952.
- [23]. H.M. Markowitz, *Portfolio selection: Efficient diversification of investments* (Yale University. Department of Economics. Cowles Foundation for Research in Economics. Monographs) ; Wiley; Chapman & Hall, 1959.
- [24]. H.M. Markowitz, *Portfolio Selection: Efficient Diversification of Investments*, Blackwell Publishers, 1991.
- [25]. D. Fylstra, "Introducing Convex and Conic Optimization for the Quantitative Finance Professional", *Wilmott Magazine* pp. 18-22, 2005.
- [26]. S. Lummer, M.W. Riepe, "Taming Your Optimisation: A Guide through the pitfalls of Mean Variance Optimisation" Presented in *Global Asset: Techniques for Optimising Portfolio Management*, Edited by J.S. Lummer, M.W. Riepe, , John Wiley & Sons, 1994a.
- [27]. S. Lummer, M.W. Riepe, "The Role of Asset Allocation in Portfolio Management", Presented in *Global Asset Allocation: Techniques for Optimising Portfolio Management*, Edited by Edited by J.S. Lummer, M.W. Riepe, John Wiley & Sons, 1994b.
- [28]. Dempster, Germano, Villaverde, "Global Asset Liability Management", *British Actuarial Journal*, Volume 9, Number 1, 2003, pp. 137-195(59).
- [29]. Althoff, K, "Case-Based Reasoning", *ICCBR-01 Workshop On Case-Based Reasoning Authoring Support Tools*, 31st July, 2001